CENTRE FOR CIVIL SOCIETY AND GOVERNANCE

THE UNIVERSITY OF HONG KONG

The social mechanisms of circular economy

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Lecture Outline

I. Adopting a circular economy framework

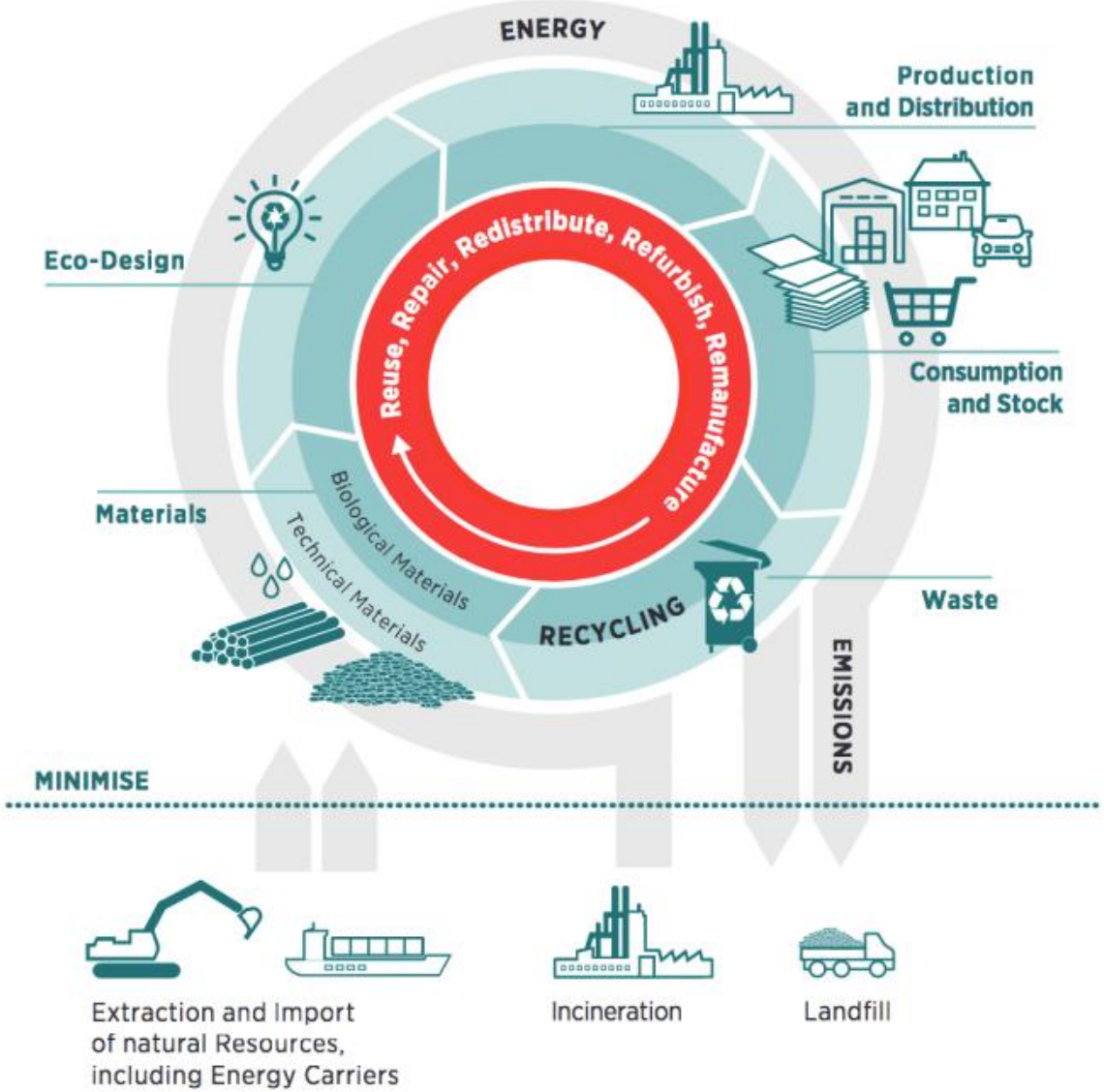
- The geographical approach: industrial symbiosis and eco-industrial parks (eco-industrial development)
- The product-oriented approach: sustainable supply chain

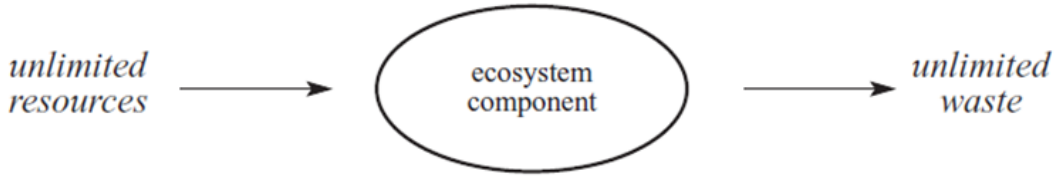
II. The social mechanisms of eco-industrial development

- Self-organisation (IS in Kalunborg, Denmark)
- Government planning (TEDA, China)
- Facilitation (NISP, the UK)

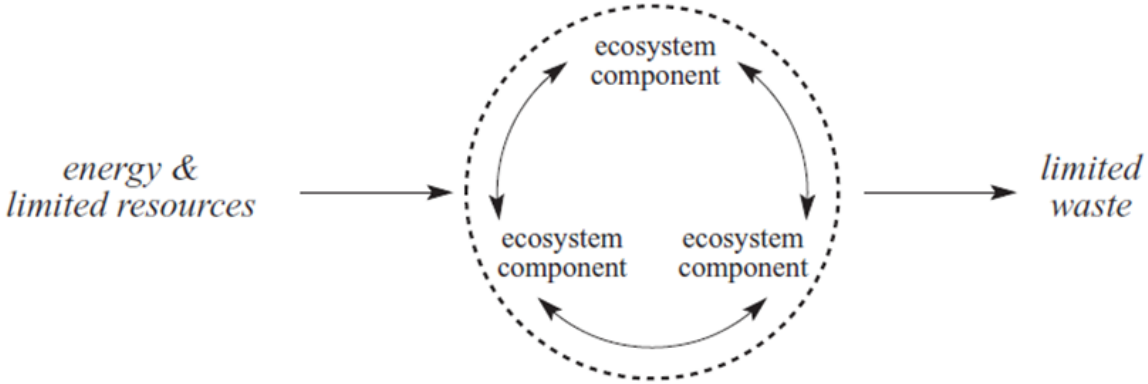
Concepts of Circular Economy

Circular economy is defined as ‘a regenerative system in which resource input and waste, emission, and energy leakage are minimized by **slowing, closing, and narrowing energy and material loops**’ (Geissdoerfer et al., 2017).



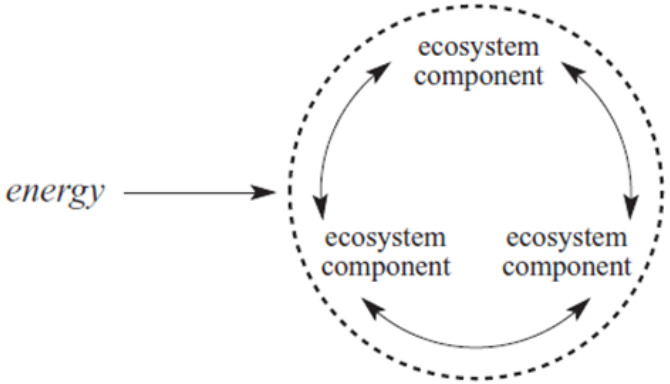


(a) QLinear materials flows in 'type I' ecology



(b) Quasi-cyclic materials flows in 'type II' ecology

The vision: *developing a nearly closed-loop industrial ecosystem by mimicking natural ecosystems*



(c) Cyclic materials flows in 'type III' ecology

Fig. 1. Typology of industrial ecosystems.
Source: Graedel and Allenby, 1995.

The product-oriented approach

The product-oriented methods (e.g. design for environment [DfE], life cycle analysis [LCA], sustainable/green supply chain management [SSCM]) **use life cycle thinking to track and reduce environmental impacts of every process of a specific product/material** (Graedel and Allenby, 1995), or employ eco-innovative techniques such as product-service system (PSS) or sustainable substitutions to replace current products (Mont, 2002).

A product-based circular economy model (O'Brien et al., 2018)

The geographical approach

The geographical approach is also known as a regional approach. It aims at **shaping the flows of materials, energy and information towards sustainability through local, regional and global economies** (Chertow, 2000).

Sustainability

↓

Industrial ecology

Facility or Firm

- design for environment
- pollution prevention
- "green" accounting

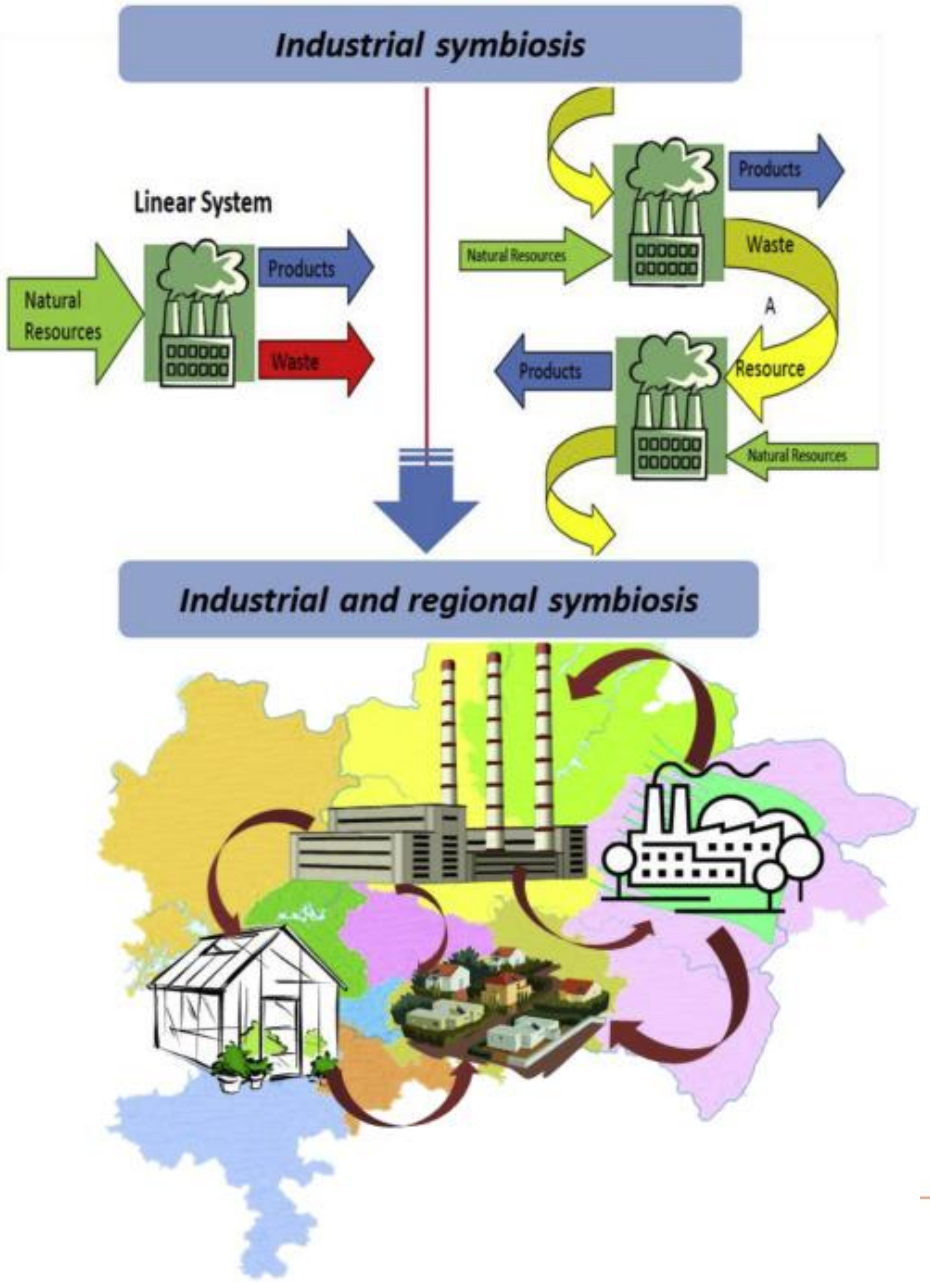
Inter-Firm

- industrial symbiosis (eco-industrial parks)
- product life-cycles
- industrial sector initiatives

Regional/Global

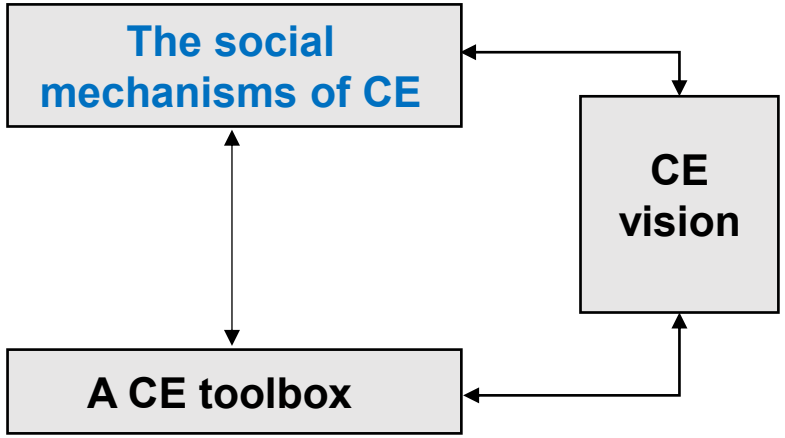
- budgets and cycles
- materials and energy flow studies (industrial metabolism)

Industrial ecology operates at three levels.
Source: Chertow (2000)

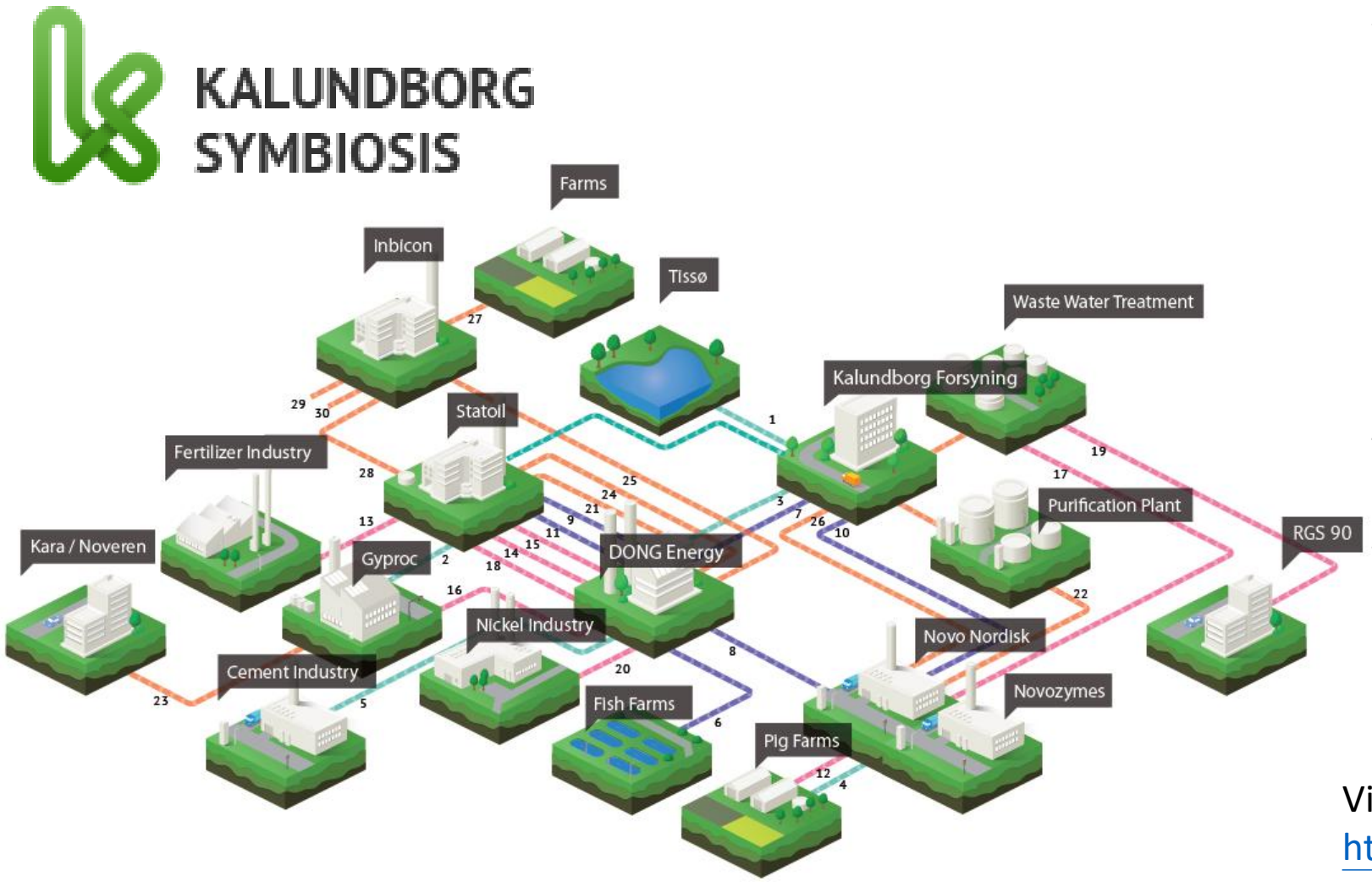


Social Mechanisms of Circular Economy

Cohen-Rosenthal (2000, p250) argues the answers to the question of "how to create connections with more value and less waste" do not lie in technologies, but "**in the ways humans make choices**, **their willingness to seek out new connections, to invent new combinations, to explore the possibilities** of the world around us."



Self-organisation: Industrial symbiosis in Kalundborg



An exchange network of materials and energy between companies and communities started in the 1960s.

Utility sharing: energy generation, wastewater treatment, etc.

Synergy initiatives:
the physical exchanges of materials, energy, water, and by-products among firms, farms, and communities

Video: Introduction to industrial symbiosis
https://www.youtube.com/watch?v=7daVvUsvBuc

Self-organisation: Industrial symbiosis in Kalundborg

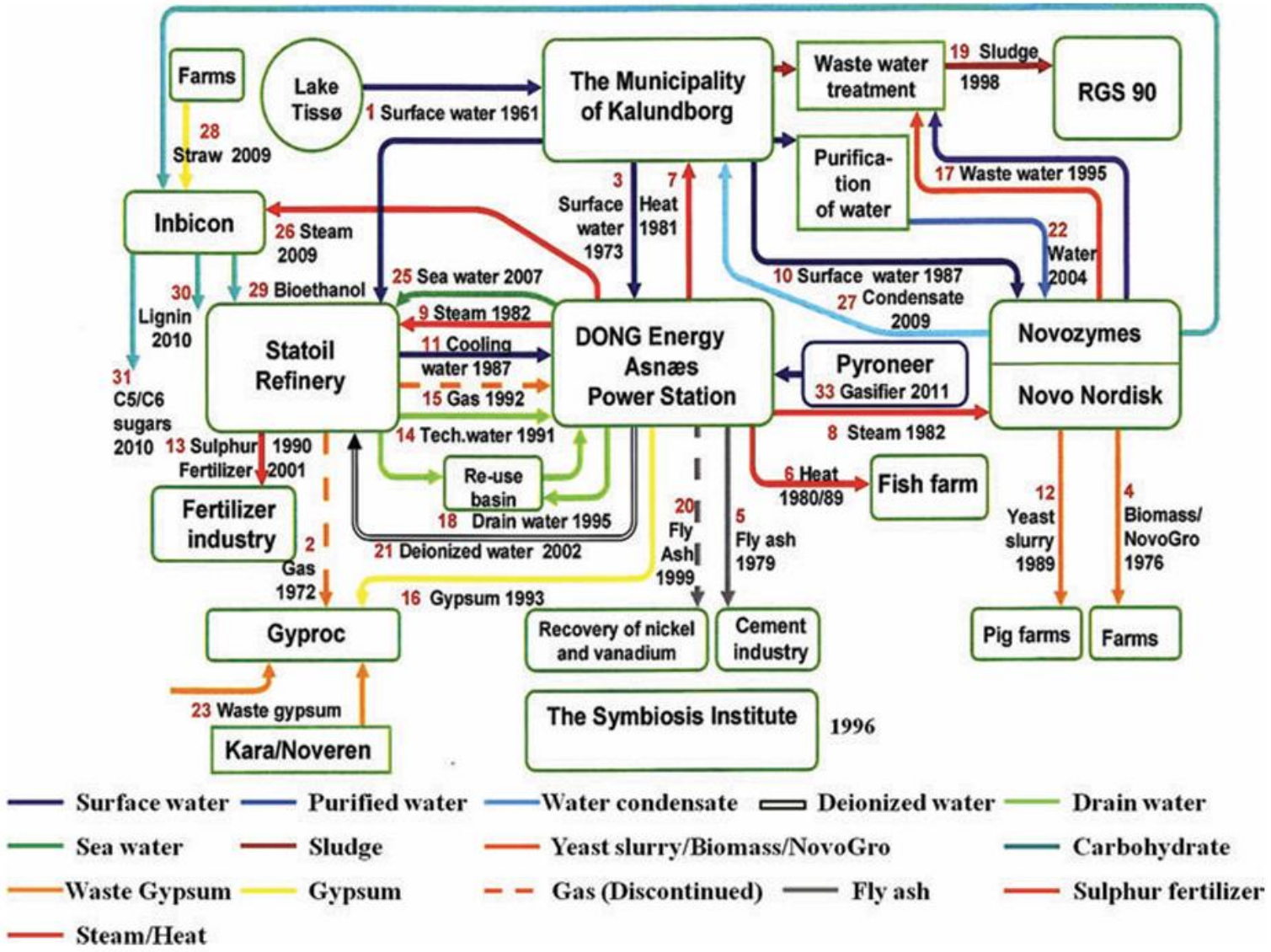


Figure 2 The Kalundborg industrial symbiosis. Actors and exchanges of materials and energy. Exchanges are numbered from 1 to 33 and the years shown indicate when an exchange began. Discontinued links are shown as dotted lines. Modified from http://www.symbiosis.dk/en/system.

- **Antecedents**: Geographical proximity
- **Drivers**: Economic efficiency and environmental benefits
- **Lubricants**: Social capital, mental and cultural proximity
- **Facilitation**: Institutionalisation

Evolution to sustainable collaboration in Kalundborg.

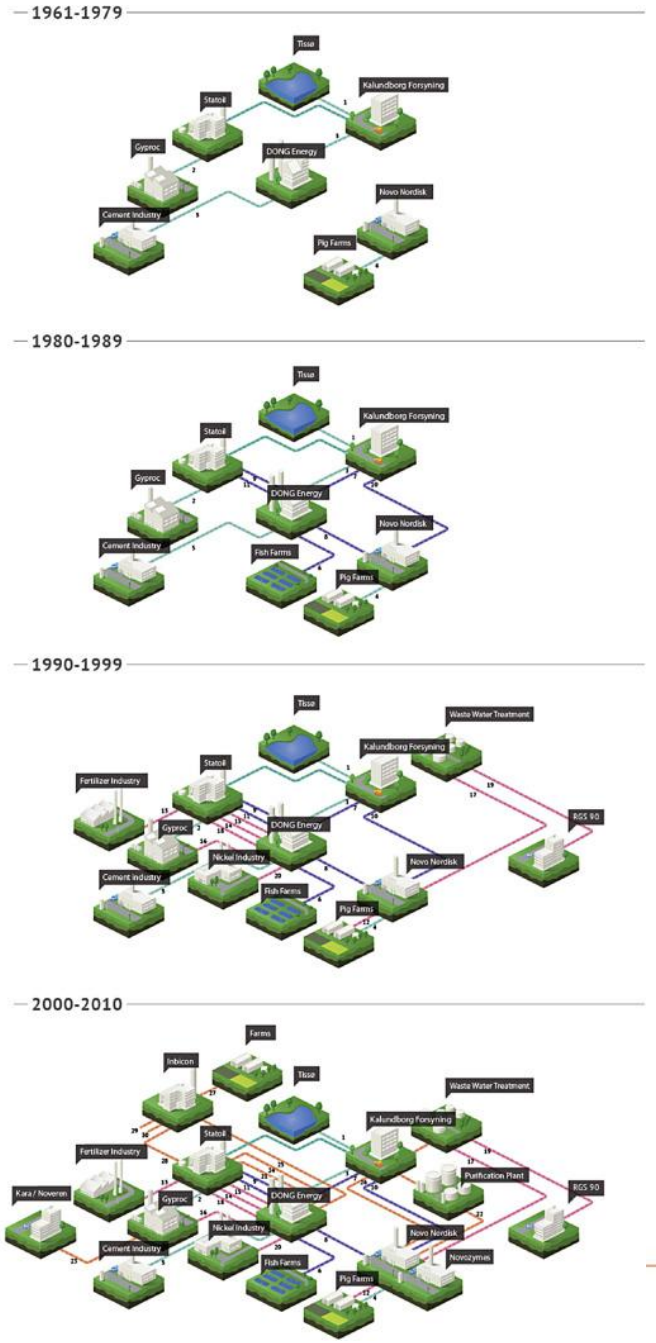
Cautious Engagement → Cooperative Escalation → Operational Collaboration

Evidence from the Kalundborg Symbiosis

All transactions based on contracts. Gyproc diversifies gypsum supply despite deliveries from DONG Energy. Statoil keeps its boiler on back up despite the steam supply contract with DONG Energy.	Statoil reciprocates DONG's steam supply by providing DONG with cooling water and gas. Symbiosis members cooperate financially in expanding conduits for sharing resource flows within the Symbiosis boundaries. Symbiosis members agree to support the establishment of a central Symbiosis Institute.	DONG Energy begins to send deionized water to Statoil Kalundborg Forsyning establishes facilities to purify water for use at Novozymes/ Novo Nordisk Novozymes supplies biomass to Inbicon (DONG subsidiary) for creating bioethanol which in turn is sent to Statoil Farmers who were taking advantage of fertilizer from Novo Nordisk begin supplying straw to Inbicon plant

Kalundborg Symbiosis
Diagram 1961-2010

1	Surface Water	*1961*
2	Gas	*1972*
3	Surface Water	*1973*
4	Biomass/NovoGro	*1976*
5	Fly Ash	1979
6	Heat	*1980/89*
7	Heat	*1981*
8	Steam	*1982*
9	Steam	*1982*
10	Surface Water	*1987*
11	Cooling Water	*1987*
12	Yeast Slurry	1989
13	Sulfur Fertilizer	1990/2001
14	Tech. Water	1991
15	Gas	1992
16	Gypsum	1993
17	Waste Water	1995
18	Drain Water	1995
19	Sludge	1998
20	Fly Ash	1999
21	Deionized Water	2002
22	Water	2004
23	Waste	2004
24	Sea Water	2007
25	Steam	2009
26	Condensate	2009
27	Straw	2009
28	Bioethanol	2010
29	Lignin	2010
30	C5/C6 sugars	2010



(source: Valentine, 2016)

Self-organisation: The role of government and policy

Kalundborg municipal government is **an active participant** in the symbiotic network:

- the municipality procured heat for the community through a contract with the power plant in 1981,
- agreed to treat wastewater and sludge from Novo Nordisk/Novozymes under a special contract in 1995
- contracted with the same firms to purify water for their processes in 2004,
- and plays an active role in the Kalundborg Symbiosis Center (Valentine, 2016)

It also plays a role of **a proactive agent of change** by policy-making

> *"We want to create sustainable growth through increased focus on innovation, new technology and better resource efficiency through symbiotic thinking. The Kalundborg municipality is playing an active role in the transition to a climate-friendly and energy-efficient community, independent of fossil energy sources with the constant focus on exploiting and testing possibilities in new green technologies*" (Damm, 2014).

desire an irresistible circular society

Demonstration Site

Kalundborg Circular Campus, Kalundborg, Denmark

"The Circular Campus aims to integrate educational institutions into the local fabric and promote collaboration across sectors and generations",
- is partly controlled and guided by the municipality's strategic plans for the city's development and expansion (Bech and Overgaard, 2024)

Government Planning: EIP programs in China and South Korea

Governmental actors **consciously plan** place-based eco-industrial development.

To implement it in the specific regional or national context, **governmental actors** formulate strategies and develop and implement plans of action **using incentives and enforcement** (Boons et al., 2017).

Government Planning: EIP programs in China and South Korea

The CE concept has been included in policies and laws:

- Institutional **pressure** (both regulatory and normative)
- **Legitimacy** for establishing symbiotic relationships among firms
- Changing public **discourse**: fostering a shared belief on handling waste and environmental concerns

	International		China	
	Theory	Practice	Focus	Practice
1960	End of pipe technology		Soviet style urban and industrial planning	
1970		Air Pollution Control Act Japan, 1968		
		Kalundborg, Denmark, 1972-		
1980	Industrial ecology, 1989 Cleaner production, 1989		Pollution control	
1990		Eco-industrial parks(EIP), USA, 1995- Eco-town, Japan, 1997 INES Mainport Project, Netherlands, 1999-		
	Urban ecosystem, David, 1997			
2000	Urban transition Bai & Imura,2000	National industrial symbiosis, UK, 2000-	EIP, China, 2001 Cleaner production law, 2001 Eco-cities, 2003	EIPs, 2001- TEDA, China-Singapore Suzhou Industrial Park… Circular Economy Pilot Zone (CEPZ), 2001-
2005		National EIP, Korea, 2005-	Energy saving and waste reduction, 2005	
	Sustainability transition Kemp & Loorbach, 2007 Urban symbiosis Rene & Fujita, 2009		Circular Economy Promotion Law, 2009 Low carbon economy, 2009	
2010		Ecologically Equipped Industrial Area (EEIA), Italy, 2010-	Guideline of CE planning, 2011	
2012			Low carbon city, 2011 Ecological Civilization, 2012	Circular Transformation of Industrial Park (CTIP), 2012-; Low-carbon industrial parks, 2013- Eco-civilization demonstration area, 2014-
2020			Carbon emission peak and carbon neutrality, 2020	Zero-carbon industrial parks, 2025-

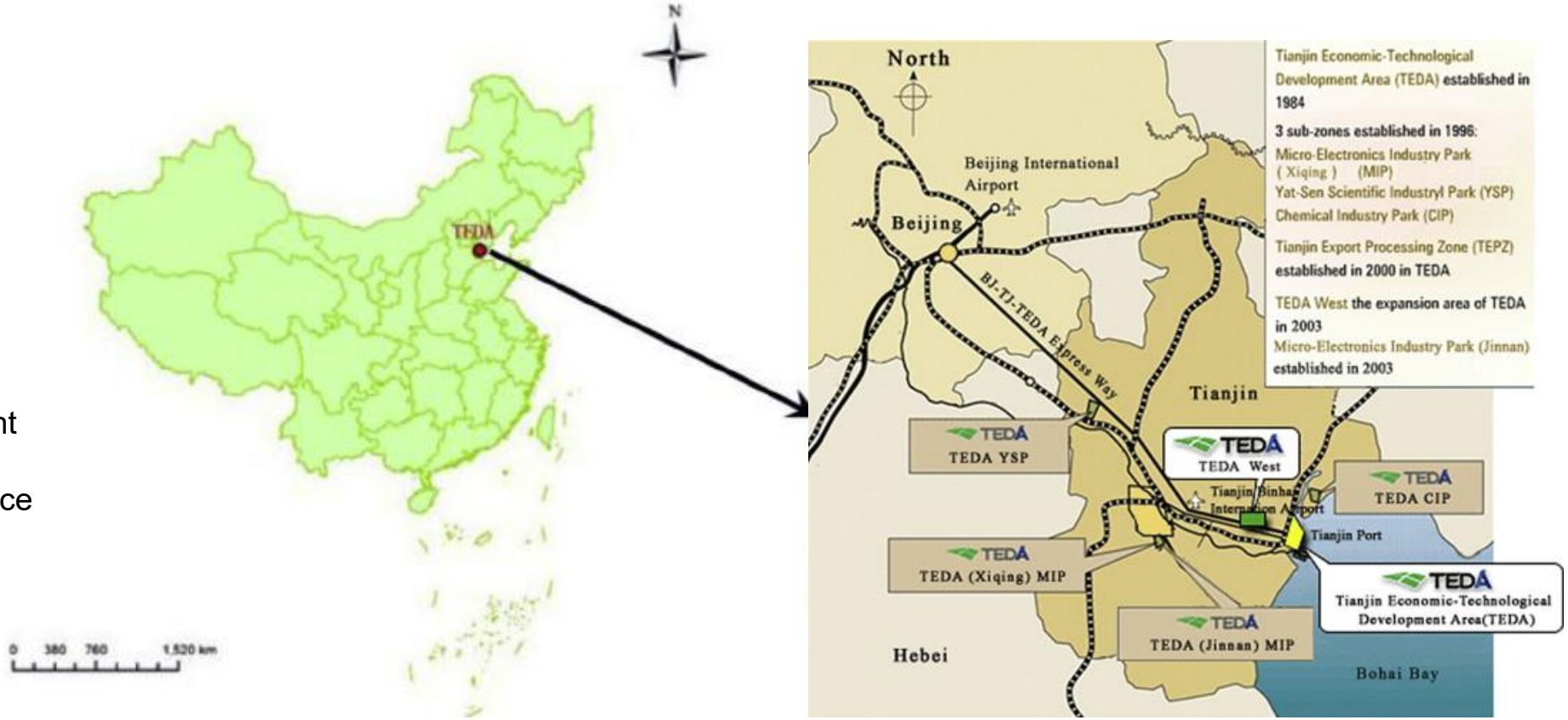
Government Planning: Eco-industrial park in Tianjin Economic-technological Development Area (TEDA)

Developed area: 98 km^2

Companies: 28,464

Main industries:
Electronics and telecommunications, automobile manufacturing, equipment manufacturing, petro-chemistry, food and beverage, biomedicine, aerospace industry, new energy and new materials.

Start time of eco-transformation: 2000



Geographical location of TEDA in China.

The initiator of the eco-transformation: **TEDA Administrative Committee**

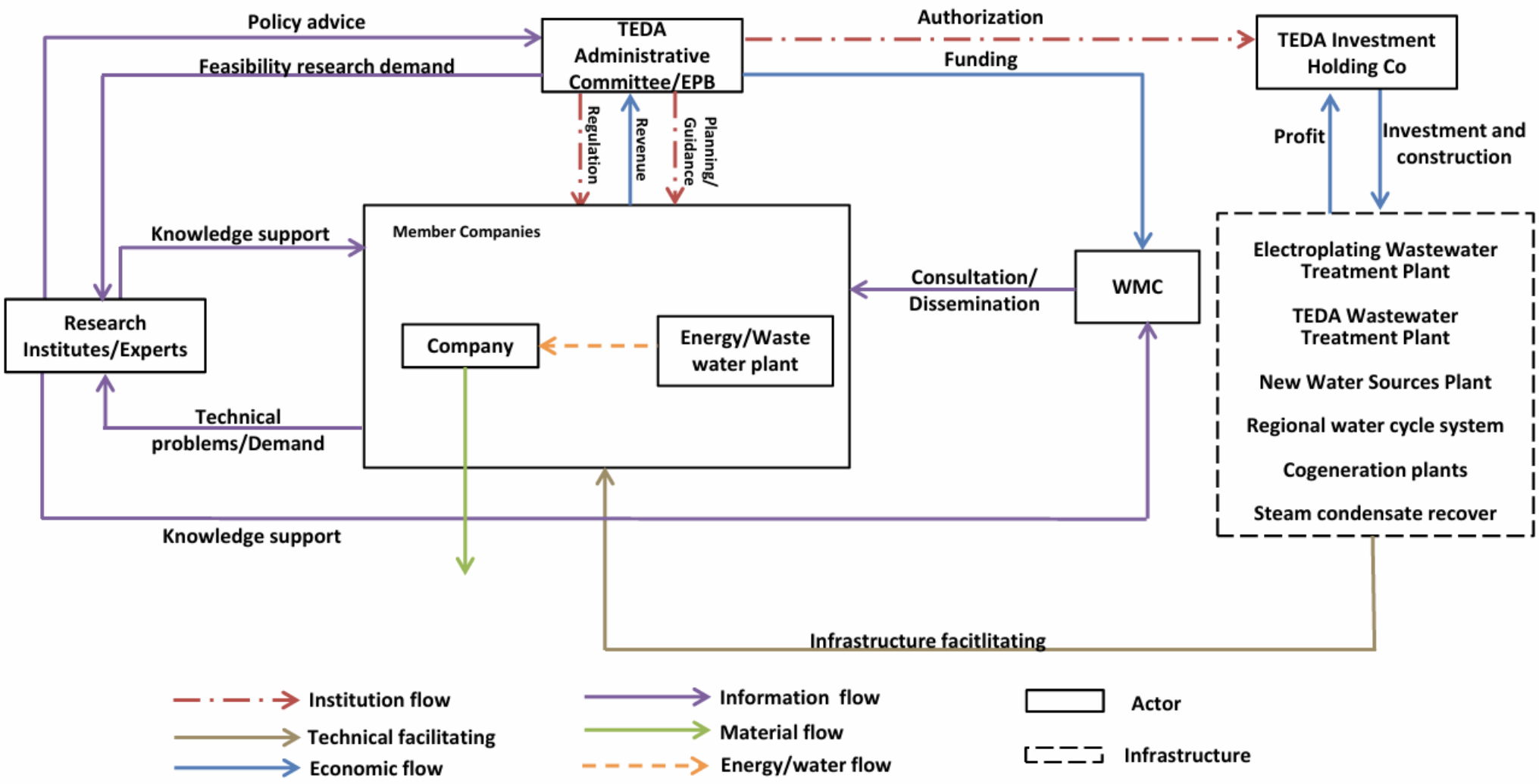


Figure 4.6 – TEDA's system structure in the initial era

The interfirm symbiotic exchange evolvement in TEDA (**at the initial stage**) (Yu, 2014).

Policy instruments to promote eco-industrial park in TEDA

Instruments/planning	Concrete implementation
Administrative organization	• Set up **leading office** for promoting circular economy. • Apply for regional ISO14000 certification.
Planning and Industrial adjustment structure	• **Preliminary assessment** by research institutes • **Engage and recruit** anchor tenants and the related supporting industries. • Recruit **complementary projects** to connect and extend the existing industrial networks. • Recruit **waste scavenger companies**.
Building/retrofitting environmental infrastructures	• **Centralized wastewater treatment and recycling**: Wastewater treatment plants; Rain water collection; Seawater desalination; Water pipelines. • **Energy system**: Cogeneration plants; Solar power and wind power for electricity and lighting; Waste-to-energy.

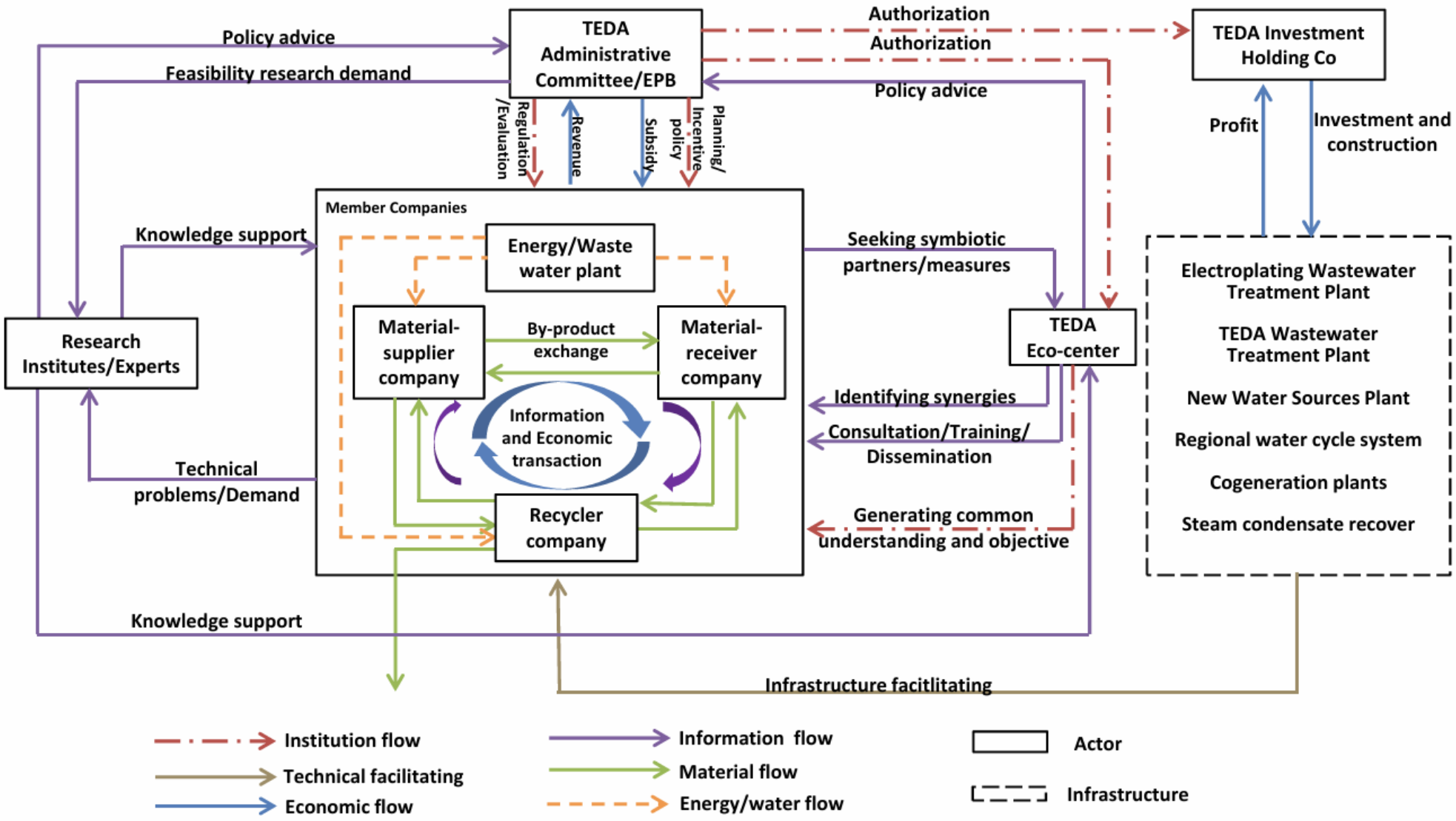


Figure 4.7 – TEDA's system structure in the emerging era

The interfirm symbiotic exchange evolvement in TEDA (**in later stages**) (Yu, 2014).

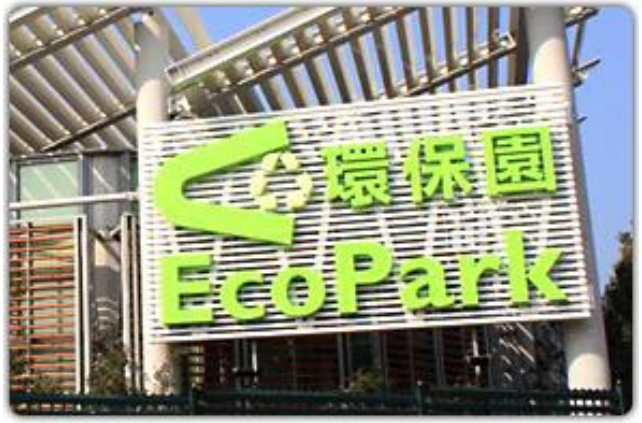
Policy instruments to promote eco-industrial park in TEDA

Instruments	Concrete implementation
Economic instruments	• **Subsidize** companies using reclaimed water. • **Fund** firms to qualify and apply for ISO14001 certification. • Fund, reward, give priority of **land supply** and discount **loans**, provide **tax exemption / deduction** for energy-saving and environmental protection projects.
Regulatory instruments	• **Mandatory** cleaner production and energy **auditing** for energy-intensive consumers or polluters. • One ballot **veto** for intensive energy/material/emission-intensive projects. • Examine and issue pollution **discharge permits,** waste treatment permits. • Carry out **administrative and economic penalties** to firms that fail environmental assessments.
Voluntary instruments	• Disseminate the **knowledge** of industrial symbiosis, circular economy and cleaner production. • Encourage environmental **information disclosure.** • **Voluntary agreement** on energy-saving signed between AC and energy intensive companies. • Set up **a coordinating organization** to organize inter-firm networking, uncover existing synergies, and foster the potential synergies. • **Ranking environmental performance** of companies. • **Training** about ISO14001, cleaner production, industrial symbiosis and circular economy.

Government Planning: for 'planned' EIP or IS projects

Benefits

- Effective to initiate and implement the EIP program
- Be able to solve urgent pollution problems



Issues of concern

- Heavily rely on government funding
- The risk that companies are not actually engaged, even though they are located nearby
- Have companies attitude towards environmental responsibility improved?
- Unintended consequences or conflicts with other policies due to inefficient coordination by authorities (e.g., tax relief for the continued by-product generation & exchange vs waste prevention) (McDowall et al., 2017; Mathews & Tan, 2016)

A “third way” ?

Facilitation: the National Industrial Symbiosis Programme (NISP) in the UK

Began as three regional pilots in 2003 in the West Midlands region and went UK national in 2005

World's first National Industrial Symbiosis Programme (more than 12,500 business members)

Aims to help businesses in various sectors and of various sizes come together to find uses for unwanted materials, aiming to divert significant waste loads from landfill and produce bottom line benefits for companies through reduced disposal costs and new commercial opportunities, by sharing assets, resources, logistics and expertise.

Source: Mirata, 2004; Paquin and Howard-Grenville, 2012.

Facilitation: the National Industrial Symbiosis Programme (NISP) in the UK

The development of industrial symbiosis networks through 'facilitated emergence', relies on the presence of **a brokering organization** to bring firms together.

Brokerage is defined as the process "by which **intermediary actors** facilitate transactions between other actors lacking access to or trust in one another" (Marsden, 1982: 202)

- 12 Regional practitioner teams (coordinators) across the UK
- 12 Programme Advisory Groups (PAGs) made up of key business people from each region, work closely with the NISP teams

Brokering activities – from hosting networking events and workshops for firms, to facilitating direct introductions between interested firms, and coordinating in-depth consulting projects with individual firms.

Facilitation: the National Industrial Symbiosis Programme (NISP) in the UK

- **Key role of the network brokers:** NISP regional coordinators
- **Government support in the initial phase**

Synergy workshops: Facilitating the exchange of information and best practice between businesses
Tried and tested, interactive business opportunity model

(Mirata 2004; Paquin and Howard-Grenville 2012).

Really simple principle: it's all about ***making matches!***

What do you **'HAVE'**?

Unwanted resources, services on offer etc.

What do you **'WANT'**?

Materials, space, services etc

- Everyone writes stuff they 'HAVE' and 'WANT' on the slips provided
- These are circulated to all tables in the room and MATCHES made!
- NISP facilitators on each table are there to drive the process

I Want @ Table No:	I Have		
Company Name	Table No	Delegate Name	Company Name
Recycling Co.			
Delegate Name			
Libby Lancaster			
I Want			
Plastic			
Quantity / Volume			
Up to 100t/week			
Location			
Southampton			
Requirements (Frequency / Duration)			
Will collect. HDPE preferred.			

I Have @ Table No:	I Want		
Company Name	Table No	Delegate Name	Company Name
A Company Ltd	*3*	*Robert Spencer*	*RS Trading Co.*
Delegate Name			
Heather Rogers			
I Have			
Wood			
Quantity / Volume			
2 tonnes /week			
Location			
Winchester			
Requirements (Frequency / Duration)			
Ongoing. Need weekly collection			

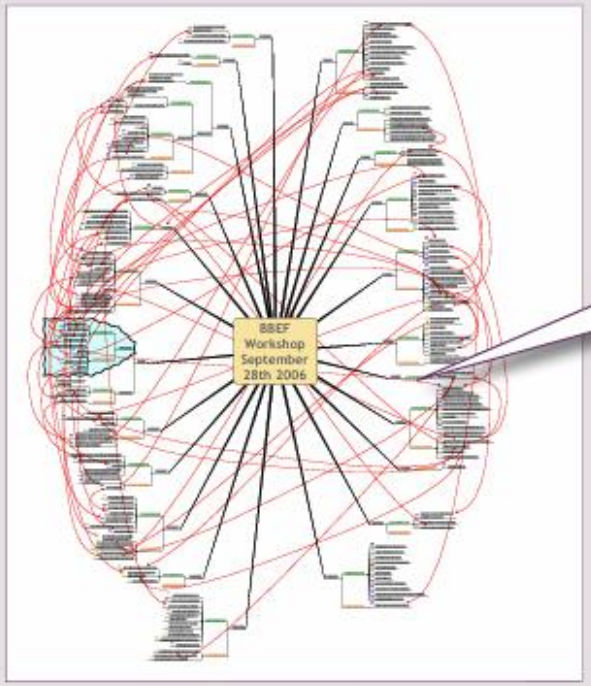
Source: NISP, 2009

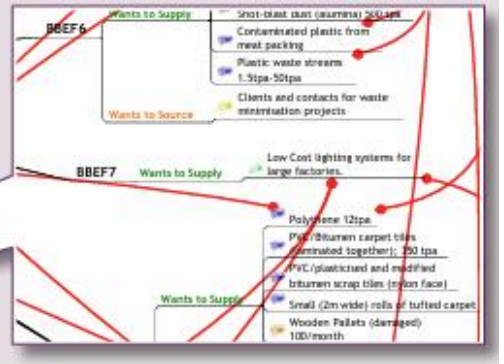
What happen next ?

Data handling: NISP Resource Database

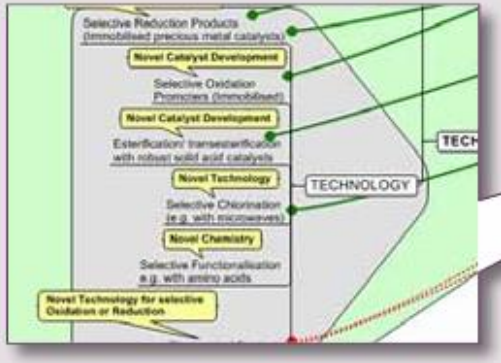
Further assistance for implementation and follow-up monitoring

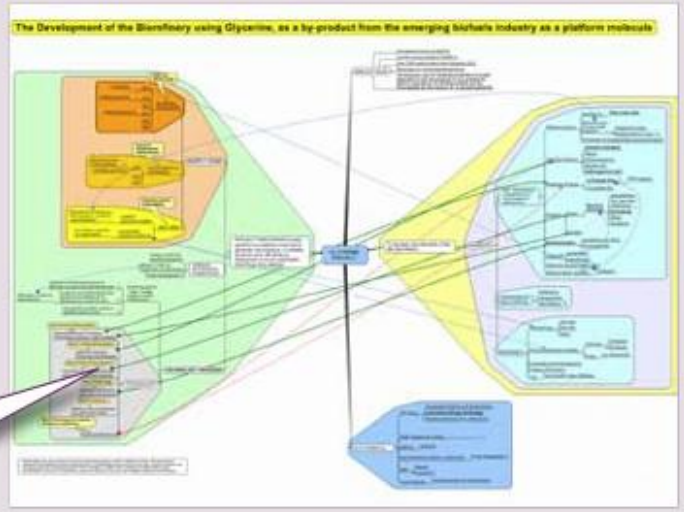
Below: Companies attending NISP interactive workshops identify their 'haves' and 'wants', the practitioners are then able to pinpoint potential connections (synergies).





Right: NISP workshops can be designed to find solutions to specific problems by bringing together producers, users and solution providers. In this case the workshop looked at identifying ways in which to recover and reuse Glycerine.





Below: NISP employs sophisticated proprietary data management and analysis tools such as CRISP.

Source: NISP, 2009

Facilitation: The role of government

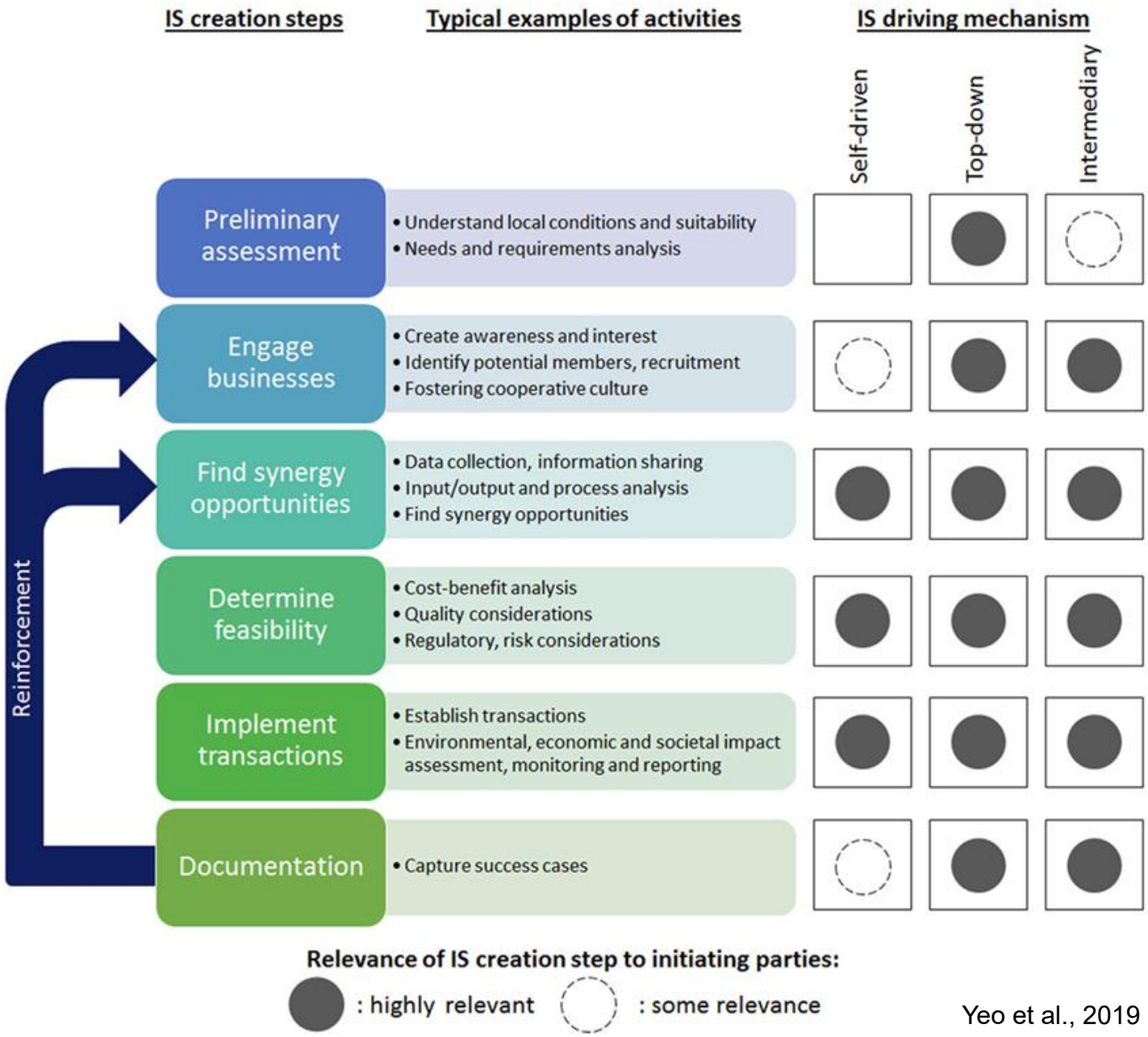
West Midlands Industrial Symbiosis Programme (since 2002/3) ← The regional government funding in the initial phase

NISP (since 2005) ← National-level funding in 2007 (£27 million)

now in transition to a self-sufficiency mode…

Takeaways!

- Eco-industrial development **can take different pathways**/modes: self-emerged, facilitated, and planned — primarily dependent on the context.
- Each pathway **can evolve and shift into other types** over time
- **Governments and policies** have roles to play in different development stages.



Yeo et al., 2019

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